Today's Agenda

1) Transformers
   → Attention is All You Need

Seq 2 Seq

Self - Attention   Context
→ Not on Proximity.

My name is Paul and I am going Londo

Sequence 2 Sequence

Encoder   Decoder

RNN :- Many to Many

Machine Translation, Text Summarization, Text Generatio

English → French

E Sentence → F Sentence

↓ ↓

Sequence of tokens   Sequence of tokens.

↓

Attention Mechanism

2022 LLM :- Large Language Models

Parameters :- 10 Billion

2018 SLM :- Small Language Models

500M ~ 10 Billion

Sequence → Tokens → Embedding → Vectors

↓

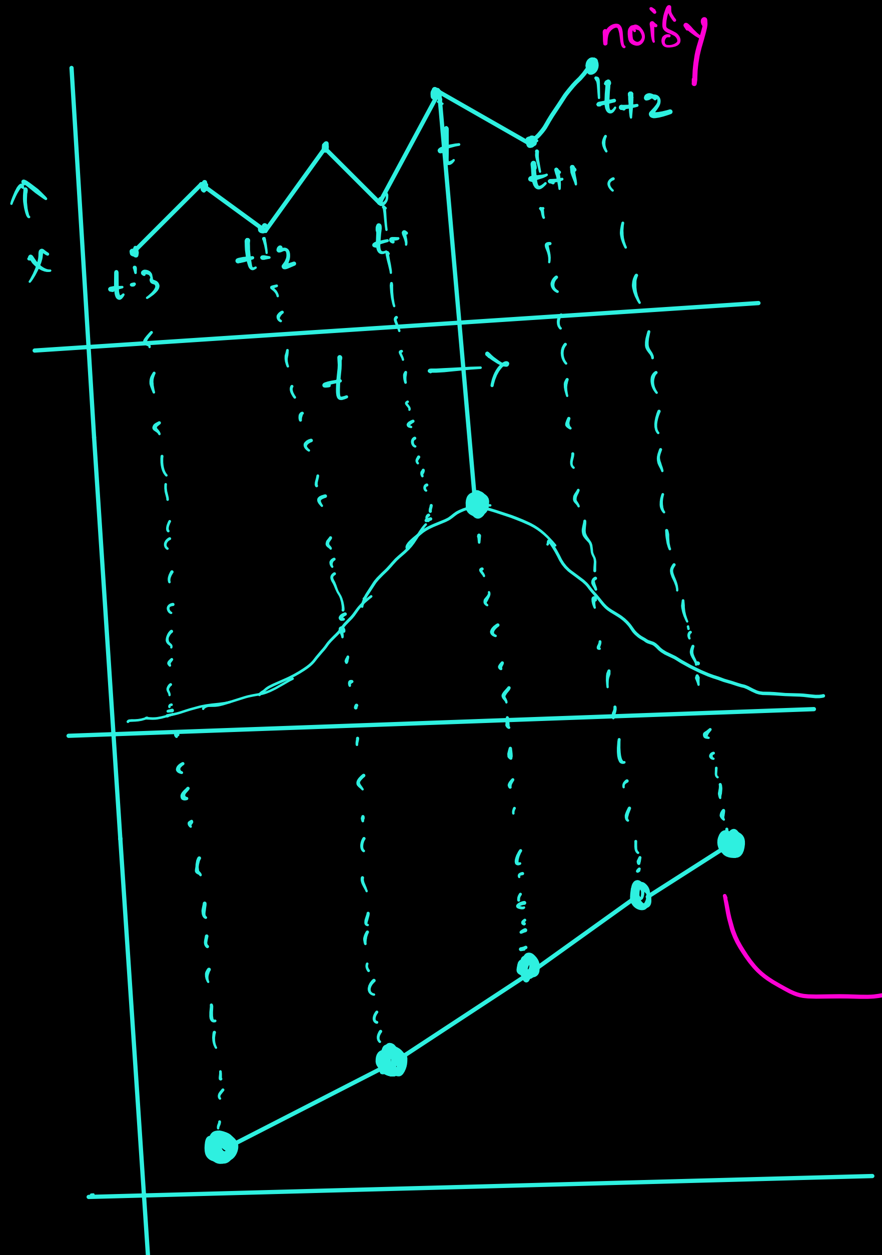
Encoding

↓

Text → Numeric

ATTENTION MECHANISMS

Time Series Data



EWA

Exponential Weighted Avg

Reweighing Scheme

$W_i = [w_0, w_1, w_2, w_3]$

1) Filter data that is far away.

2) Amplify if something is close!

Useful data points gave context with each another.

Focus on the Reweighing Scheme.

Noa can be annoying but she is a quiet cat.

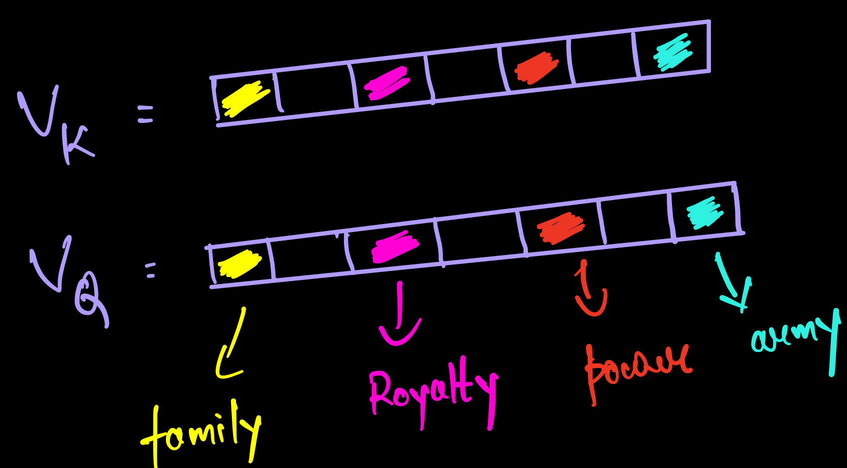
t_1 t_2 t_3 t_4 t_5 t_6 t_7 t_8 t_9 t_{10}

V_1 V_2 V_3 V_4 V_5 V_6 V_7 V_8 V_9 V_{10}

W_1 W_2 W_3 W_4 W_5 W_6 W_7 W_8 W_9 W_{10}

Y_1 Y_2 Y_3 Y_4 Y_5 Y_6 Y_7 Y_8 Y_9 Y_{10}

More context



Similar Words

Son
daughter
army
Kingdom

country
land

Dog, cat, computer
Swimming

Reweighing Scheme/Plan

$$W_{KQ} = V_Q \cdot V_K$$
$$= V_K \cdot V_Q$$

→ Explore

Today's Agenda

SELF ATTENTION

Bank of the river.

$$\begin{array}{cccc}
\text{Bank} & \text{of} & \text{the} & \text{river.} \\
\downarrow & \downarrow & \downarrow & \downarrow \\
t_1 & t_2 & t_3 & t_4 \\
\downarrow & \downarrow & \downarrow & \downarrow \\
V_1 & V_2 & V_3 & V_4 \\
\downarrow & \downarrow & \downarrow & \downarrow
\end{array}$$

$$\boxed{\text{REWEIGH}}$$

$Y_1 \quad Y_2 \quad Y_3 \quad Y_4 \leftarrow$ Better Contextualized Vectors

$V_1 V_1 = W_{11}$

$V_1 V_2 = W_{12} \xrightarrow{\text{Norm}}$ W_{11}, W_{12}, W_{13}, W_{14}

$V_1 V_3 = W_{13}$

$V_1 V_4 = W_{14}$

$\hookrightarrow$ Sum to $\underline{1}$

$$W_{11} V_1 + W_{12} V_2 + W_{13} V_3 + W_{14} V_4 = Y_1$$

Reweighing all vectors towards V_1

$$W_{21} V_1 + W_{22} V_2 + W_{23} V_3 + W_{24} V_4 = Y_2$$

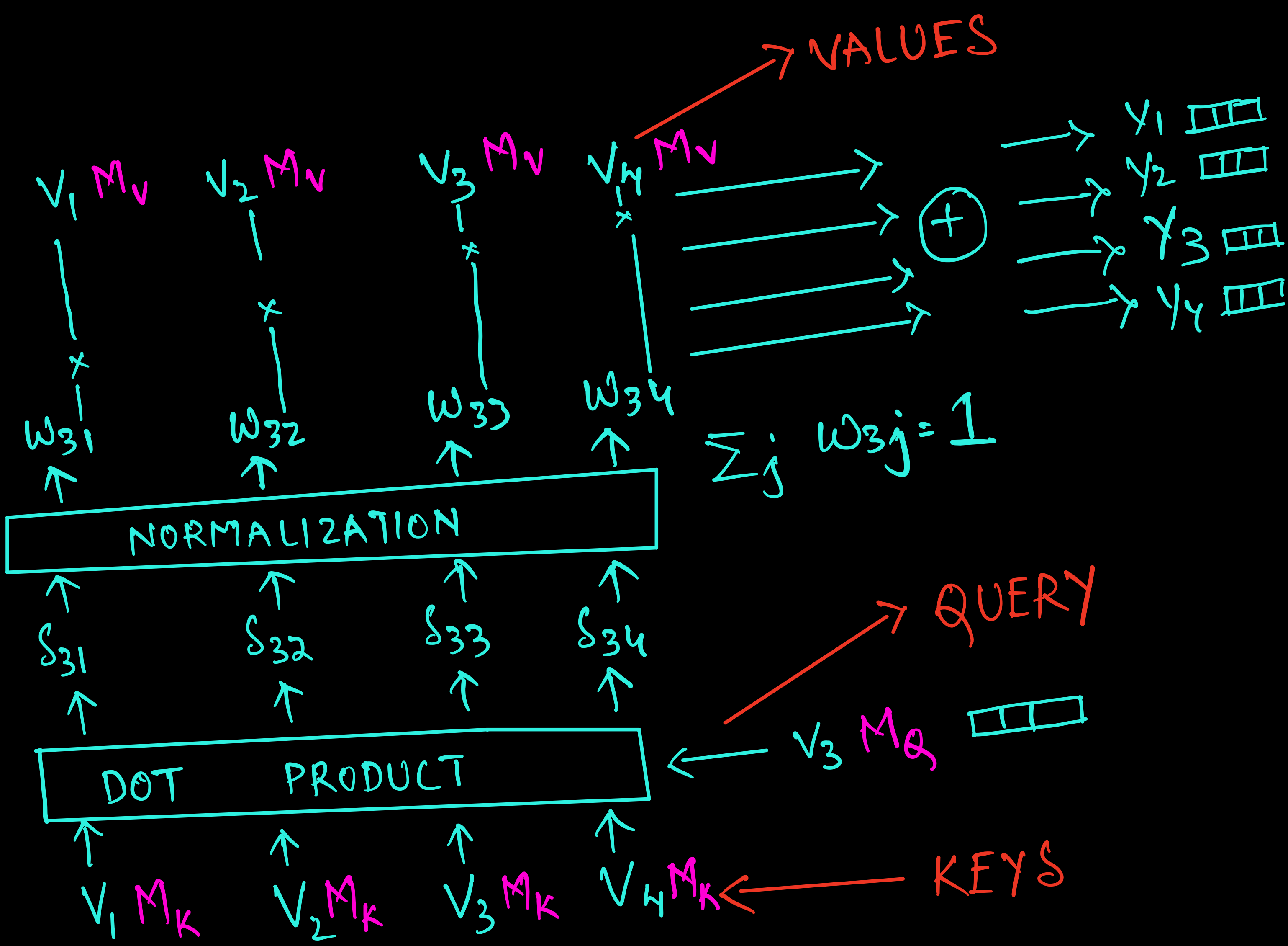
$$W_{31} \cdot V_1 + W_{32} \cdot V_2 + W_{33} \cdot V_3 + W_{34} \cdot V_4 = Y_3$$

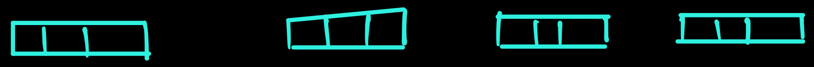
$$W_{41} \cdot V_1 + W_{42} \cdot V_2 + W_{43} \cdot V_3 + W_{44} \cdot V_4 = Y_4$$

Benefits of the Reweighing Scheme

1. I have not trained any weights
2. Order has no influence
3. Proximity has no influence
4. Shape Independent

SELF - ATTENTION

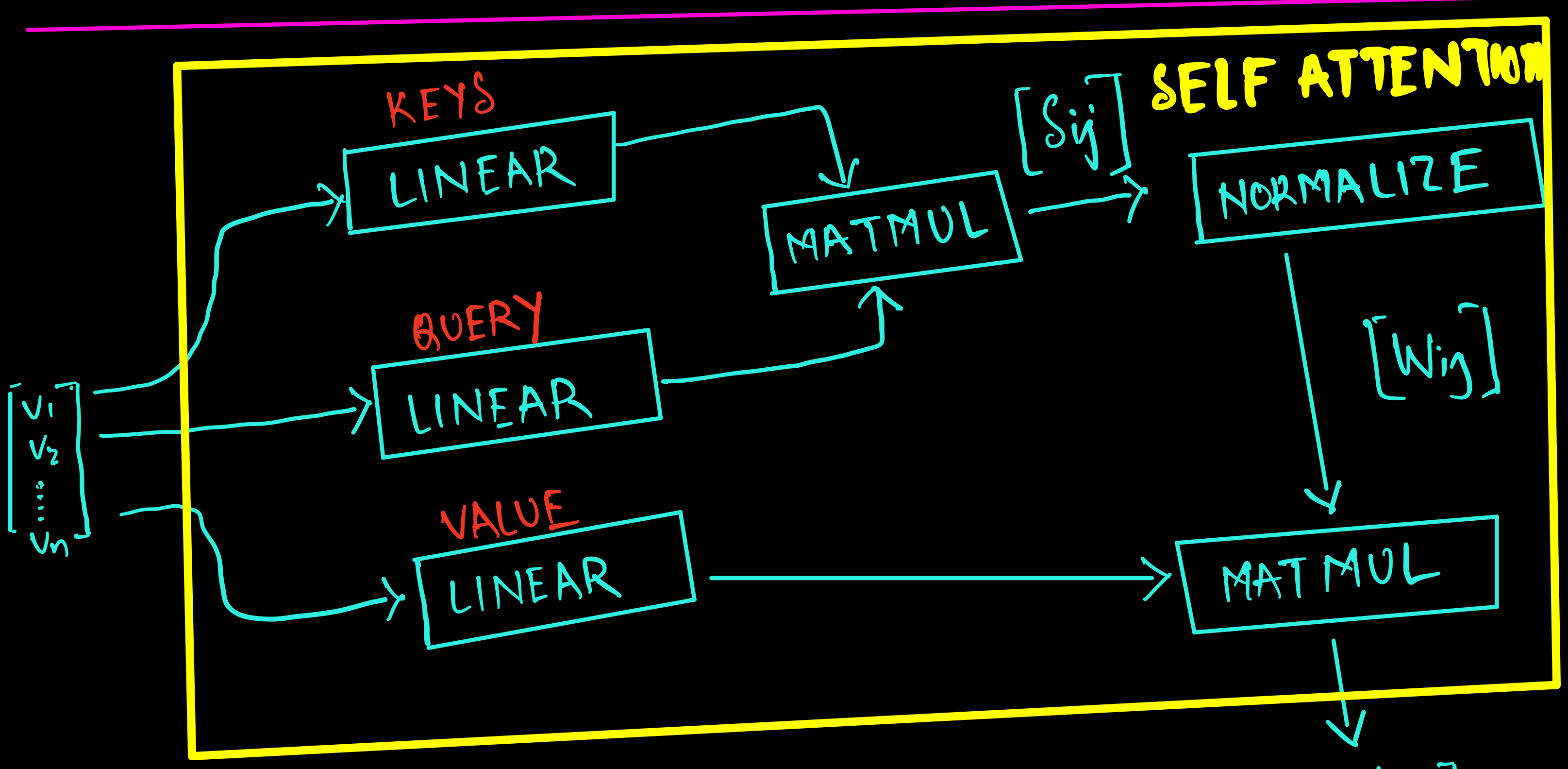


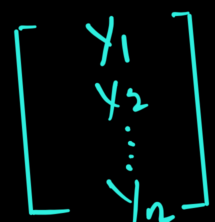


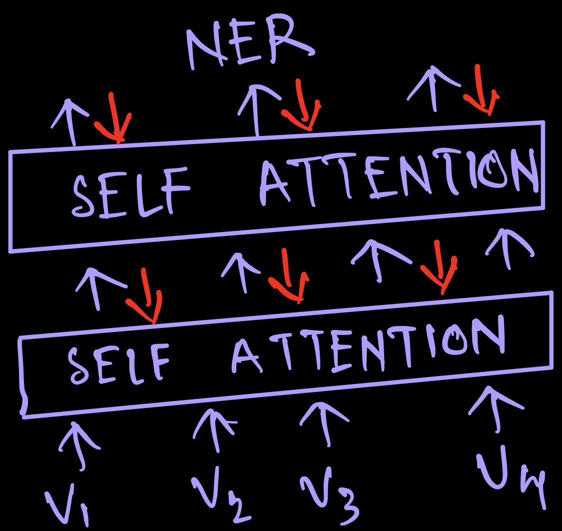
$$V_1 = (1 \times K)$$

$$M = (K \times K)$$

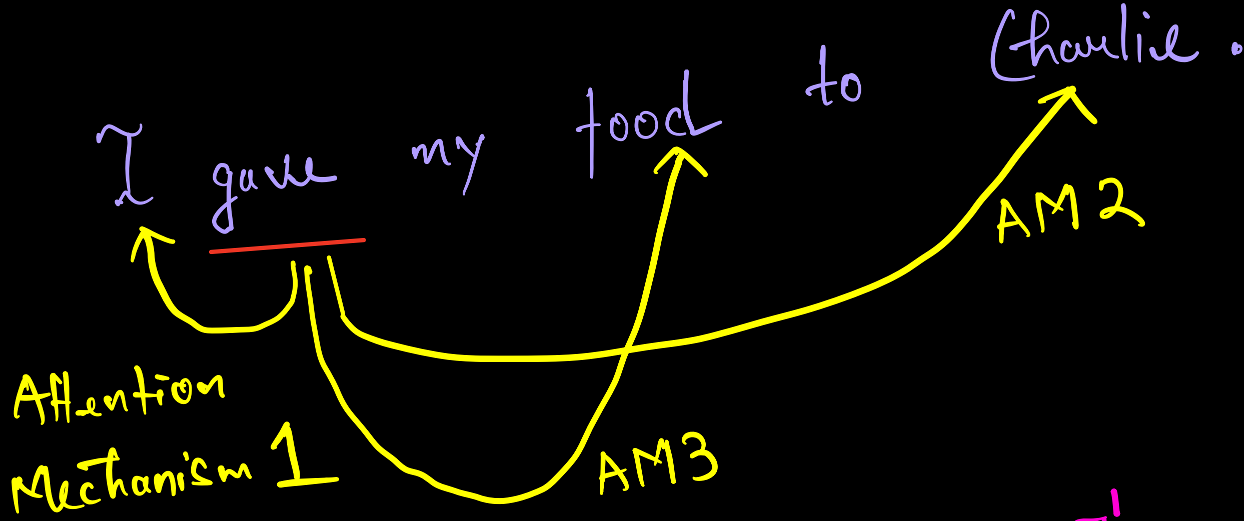
$$V_1 \cdot M = (1 \times K)$$

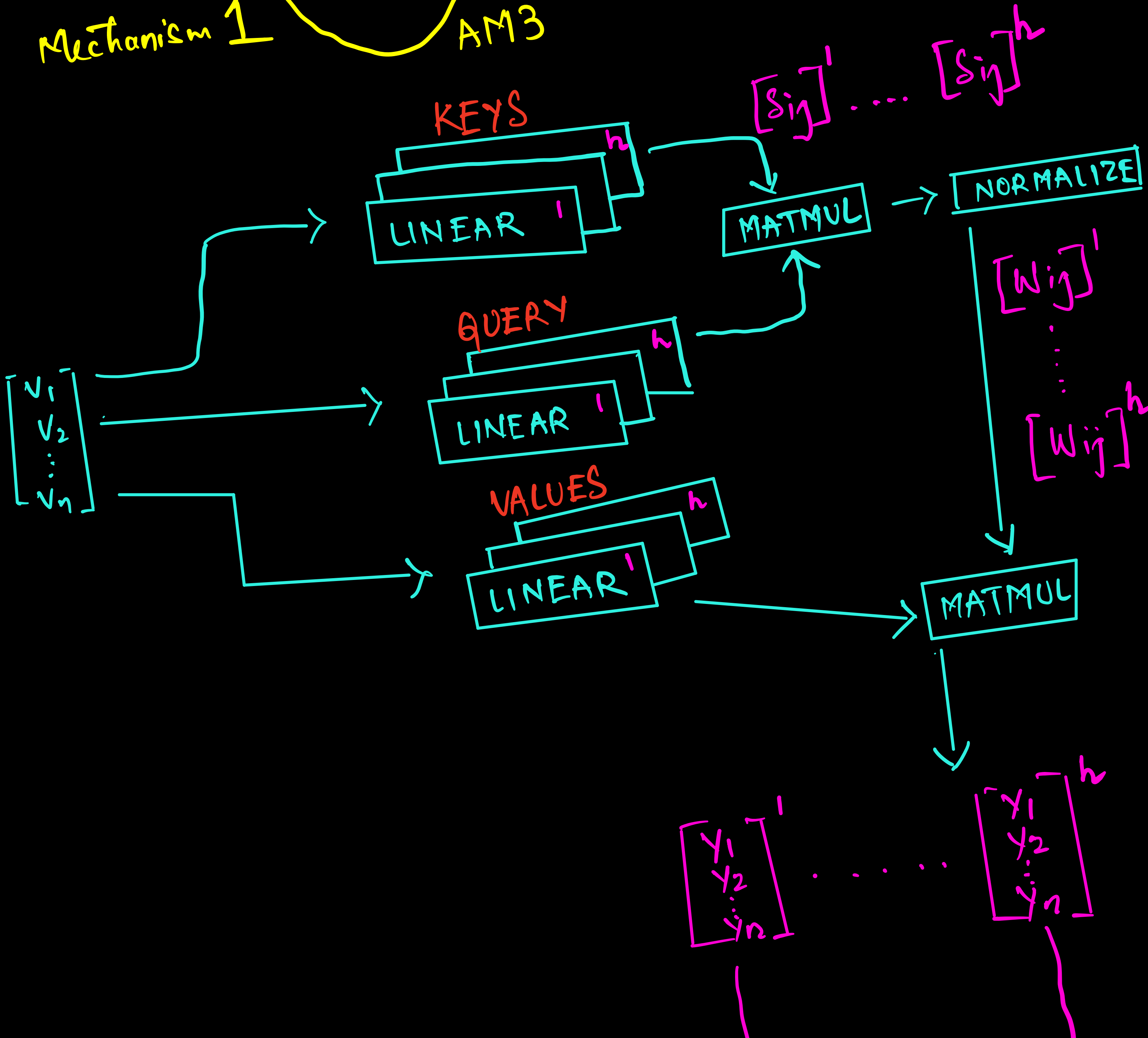




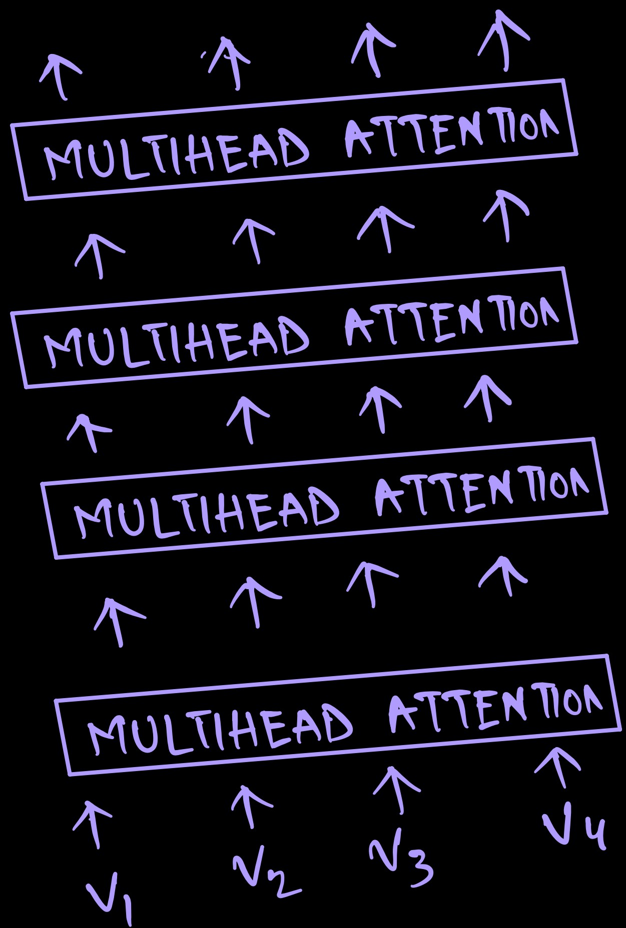


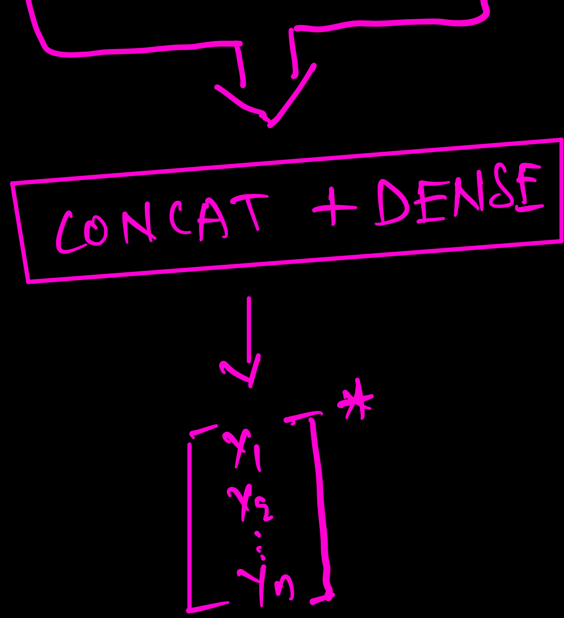
Do we have enough attention?

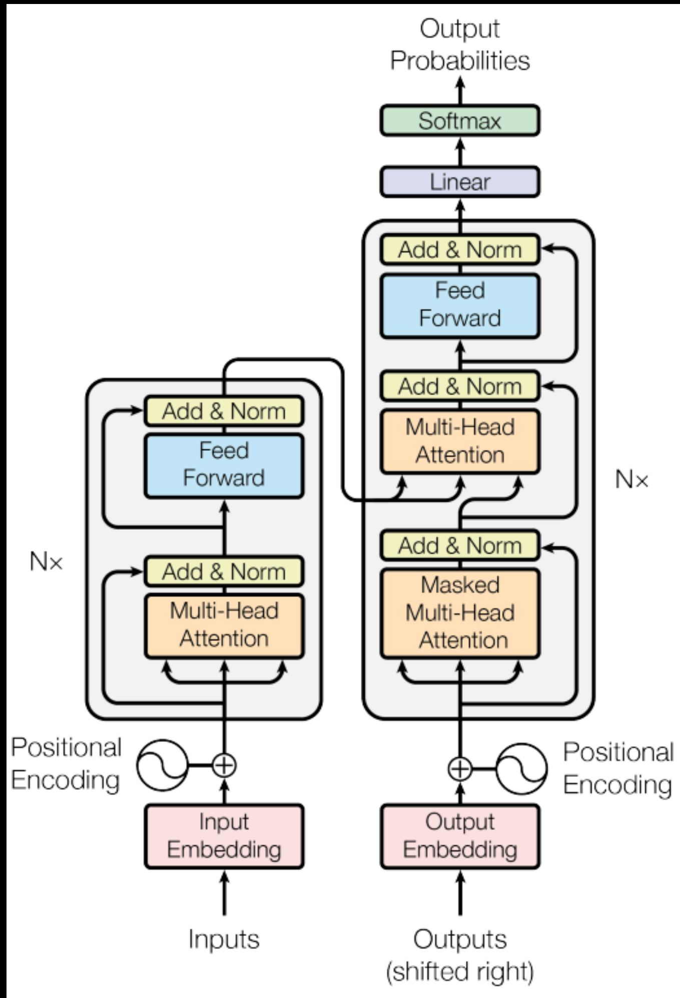




MULTIHEAD ATTENTION



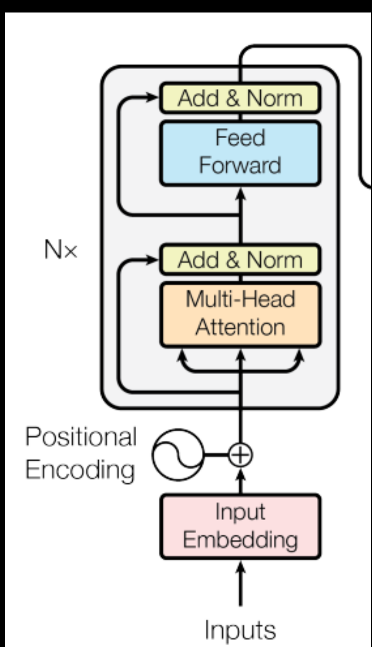




Encoder

Order has influence

Positions



P.E → Act Embedding

+

P(E,V)

⇓

(sin, cos)

Final Encoding

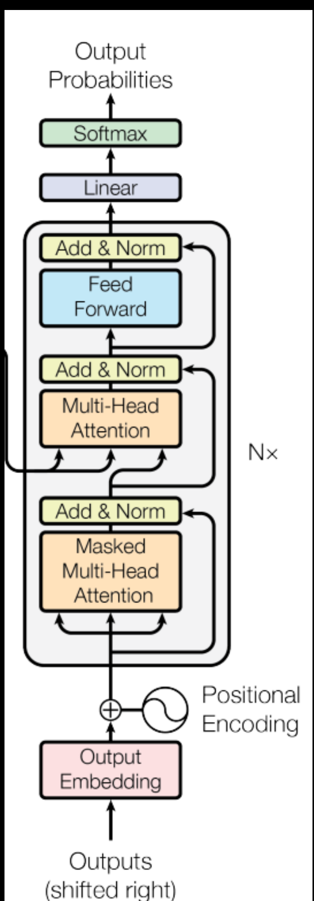
Vanishing gradient

Feed Forward Network

Linear activation function
dropout
linear

Non linearity

DECODER



Mask
↳ Hide

Standard M.H.A

My cat sat on the floor.

MASK . M.H.A

My cat 0 0 0 0

itself and past

* Nothing in future *

MASK · M·H·A

A·M

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{Q \cdot K^T}{\sqrt{d^t}}\right) V$$

M·A·M

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\left(\frac{Q \cdot K^T}{\sqrt{d^t}}\right) + \underbrace{M}_{\text{Mask Matrix}}\right) V$$

Mask

Casual Mask

mask = 4×4

= [

$$\begin{bmatrix} 0 & -\inf & -\inf & -\inf \\ 0 & 0 & -\inf & -\inf \\ 0 & 0 & 0 & -\inf \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} \text{\# Token 1} \\ \text{\# Token 1 \& 2} \\ \\ \\ \end{array}$$

]

$$\text{Softmax}(-\inf) \approx 0$$

$$PE_{(pos,2i)} = sin(pos/10000^{2i/d_{\text{model}}}) \rightarrow \text{Even}$$
$$PE_{(pos,2i+1)} = cos(pos/10000^{2i/d_{\text{model}}}) \rightarrow \text{Odd}$$

i = dimension index

512

d_{model} = Embedding length

pos = position of token in a sequence.

Why? Periodicity

Constrained Values.